

Reinformatizing Autoformalizations

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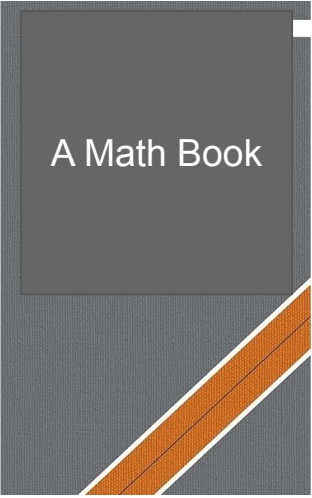
Informath Felix

NextReason
MALINCA



European Research Council
Established by the European Commission

Context:
autoformalization and informalization



A Math Book

LLM autoformalization

```
t0 a (t0 b c)) (t0 (t0 a b) c)) => match_Aop a1
a2 (plus Type2 (nd 1)) (___ : Aop a1 a2 => univ
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: a1 -> a1) => p0 : Eq t0 x01 => prod Set Prop
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```

symbolic
informalization

Thm03. $\mathbb{Q}$ is denumerable.

Thm03a.

$$|N| = |\mathbb{Q}|.$$

Thm04. Let u and v be vectors. Then $u \perp v$ implies $\|u + v\| = \sqrt{\|u\|^2 + \|v\|^2}$.

Thm07. Let $p, q \in \mathbb{N}$. Then if p and q are prime, then

$$\binom{p}{q} \binom{q}{p} = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

Thm09. Let c be a circle. Let r be a real number. Assume that $r = \text{radius}(c)$. Then the area of c is equal to $\pi \times r^2$.

Thm10FermatLittle. Let p be a natural number. Assume that p is prime. Let a be an integer. Then there exists an integer q , such that $a^p - a = p \times q$.

Thm11. $p \geq n$ and p is prime for a natural number p for all natural numbers n .

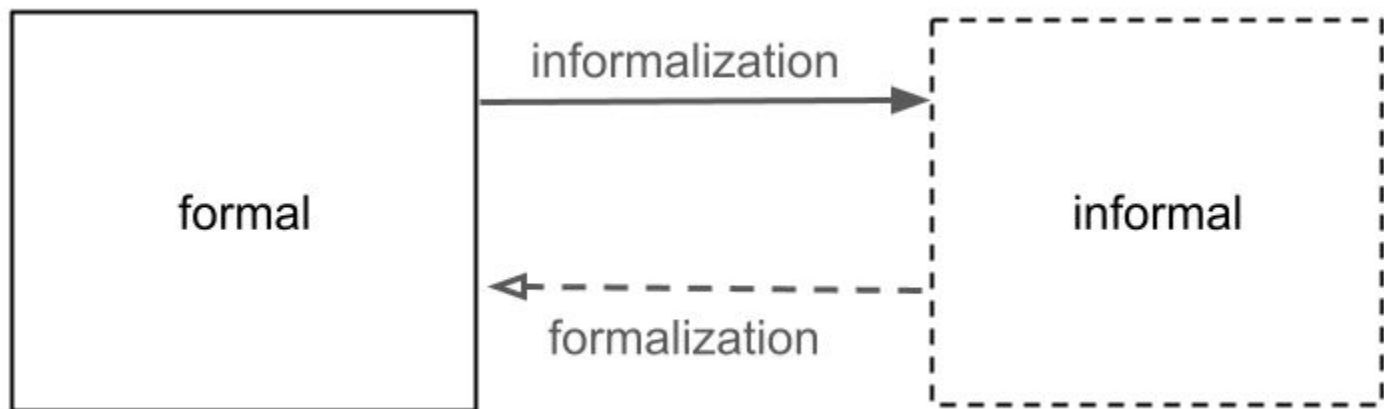
Thm19. $n = a^2 + b^2 + c^2 + d^2$ for some natural numbers a, b, c and d for all natural numbers n .

Thm20a. Let p be a natural number. Assume that p is prime. Let k be a natural number. Assume that $p = 4 \times k + 1$. Then there exists a natural number x , such that there exists a natural number y , such that $p = x^2 + y^2$.

Thm20b. Let p be a natural number. Assume that p is prime. Assume that $p \equiv 1 \pmod{4}$. Then there exists a natural number x , such that there exists a natural number y , such that $p = x^2 + y^2$.

Thm22. It is not the case that $\mathbb{R}$ is denumerable.

Mock-up example: the code and the text do not correspond to each other.



The project Informath: goals

Semantic completeness: everything expressible in type theory can be translated to Informath natural language.

Semantic soundness: everything expressible in Informath can be translated to type theory.

Linguistic completeness: everything attested in mathematical texts should be eventually covered by Informath.

Linguistic soundness: nothing should be added to Informath unless it is attested in mathematical texts.

A Math Book

LLM autoformalization

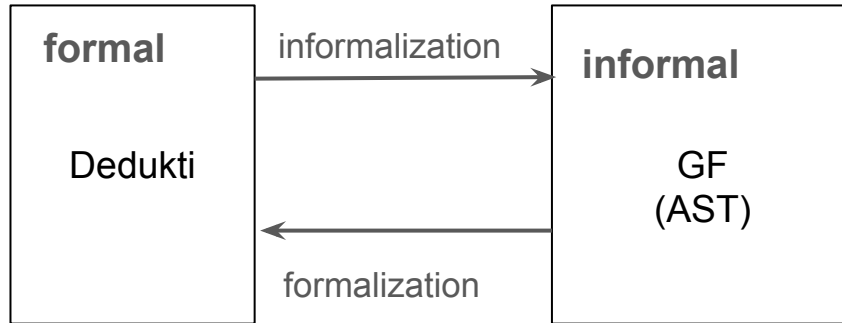
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symbolic
informalization

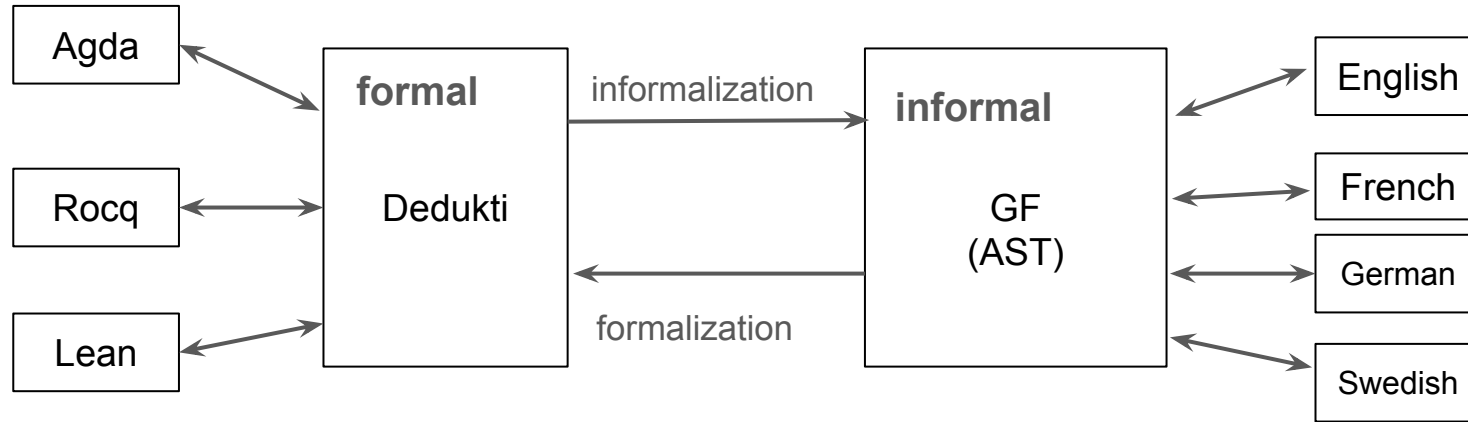
Introduction to Homotopy Type Theory

EGBERT RIJKE

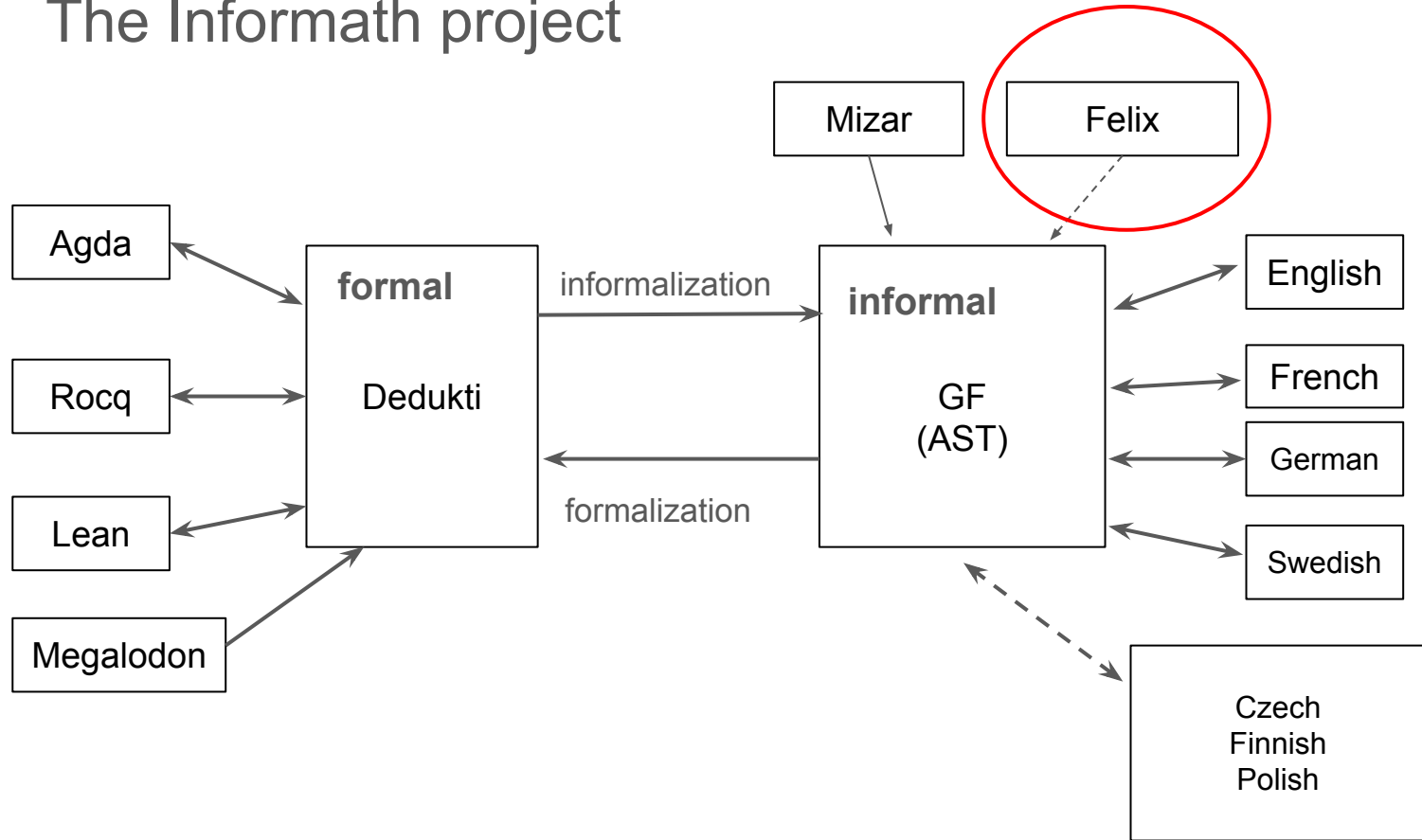
The Informath project



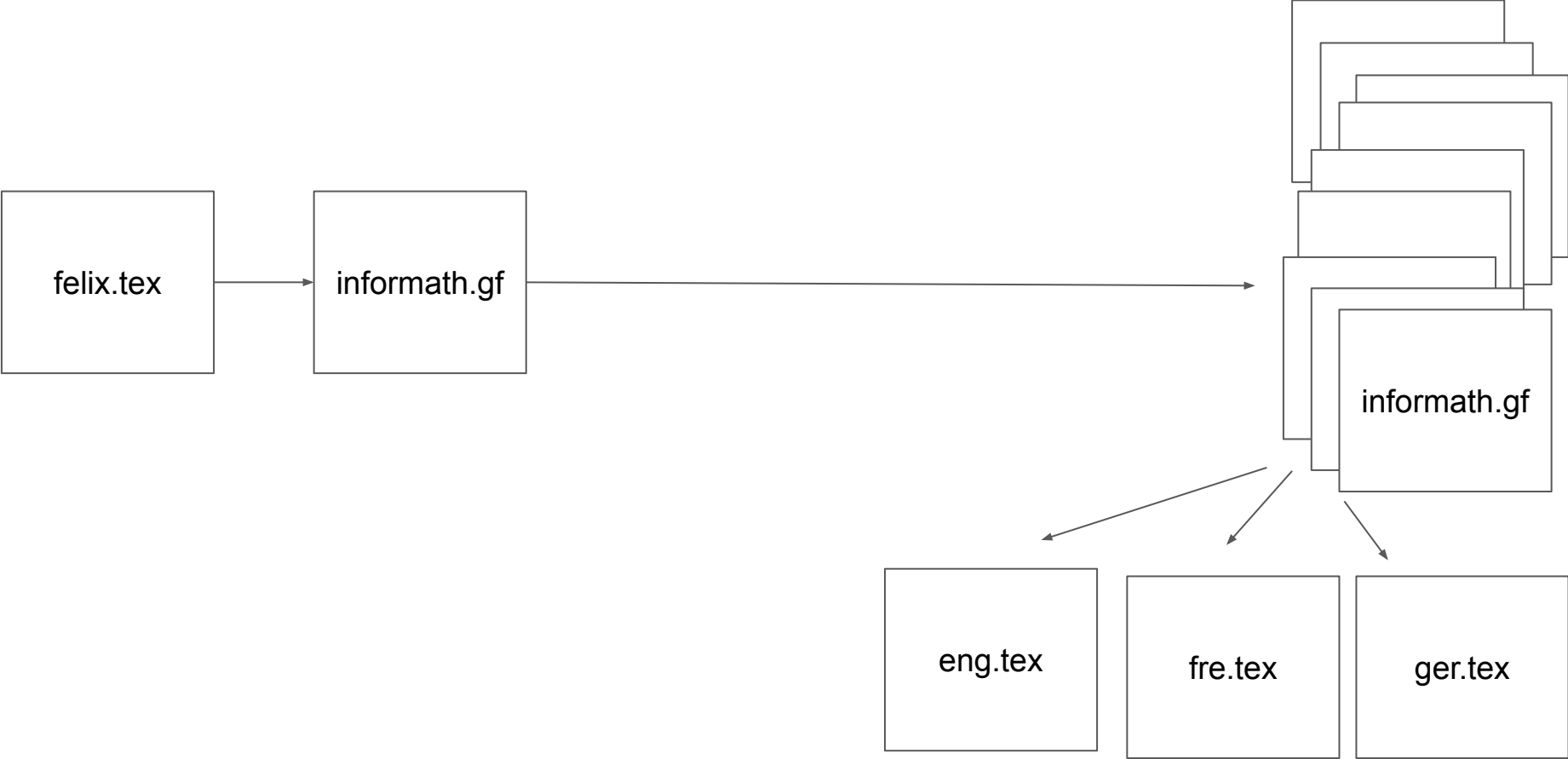
The Informath project

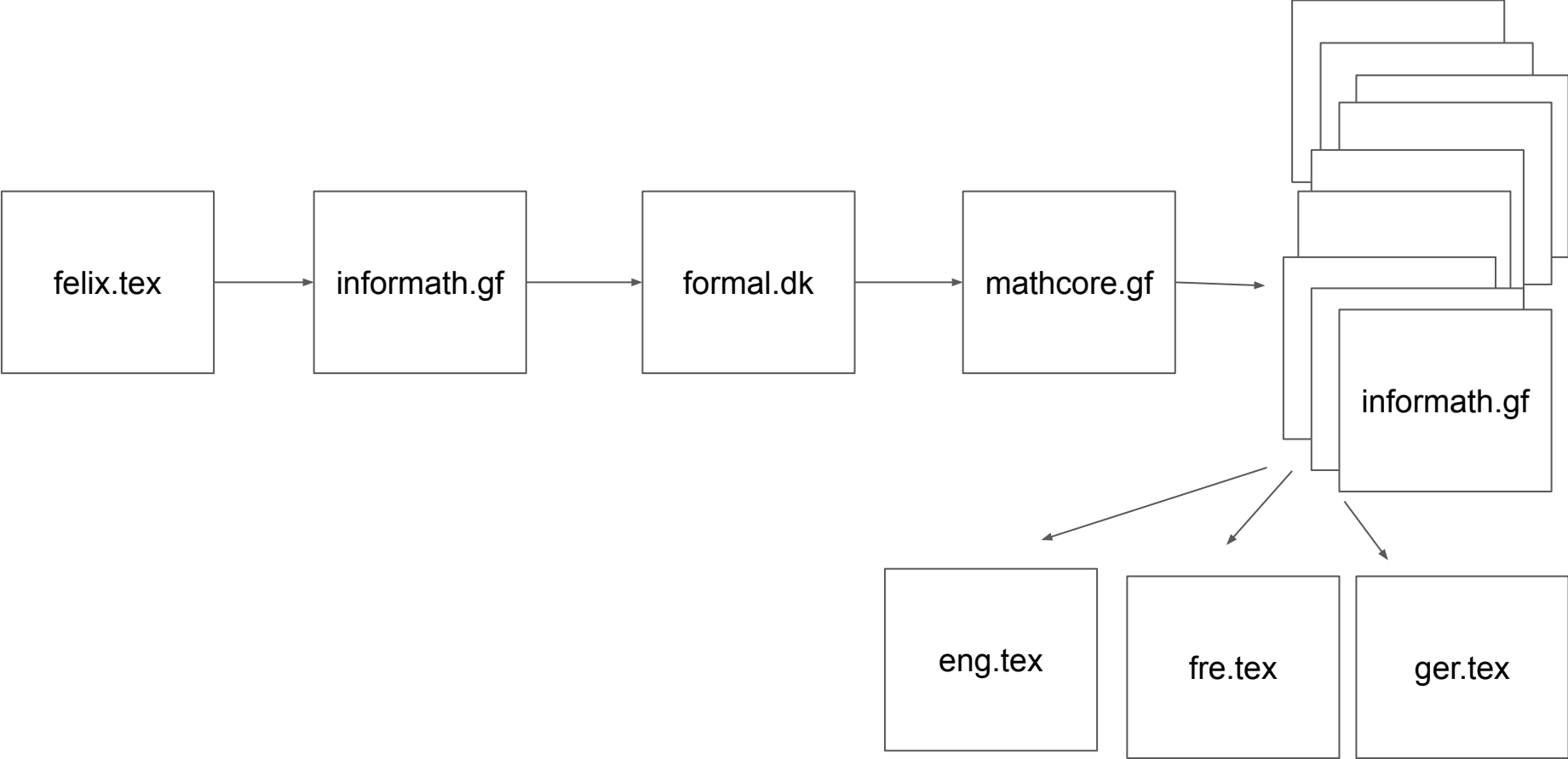


The Informath project



	Felix	Informath
functionality	proof assistant	translator
primary direction	formalization	informalization
coverage	CNL	CNL + linguistic completeness
grammar	hand-written	GF RGL
languages	English	English, French, German, ...
ambiguities	blocked	recognized
logic	HOL + set theory	dependent type theory
back end	ATP	Dedukti
proofs	scripts	proof objects





Demo

Conclusion

Status

Done:

- vibe-code conversion from Felix AST to Informath AST
- integration with Informath pipeline
 - → NLG variations via Dedukti

To do:

- conversion of proofs
- dealing with ambiguities
 - *every number is not divisible by some number*

Thanks